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United States Patent	12411451
Kind Code	B2
Date of Patent	September 09, 2025
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Image forming apparatus

Abstract

An image forming apparatus includes an apparatus body, a fixing unit attachable to the apparatus body in a first direction, and a transfer unit serving as a belt unit attachable in an attachment position within the apparatus body in a second direction intersecting the first direction. The transfer unit has a protrusion serving as a restricting portion that restricts the attachment of the fixing unit when the transfer unit is positioned in a first irregular position different from the attachment position.

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Appl. No.:	18/504584
Filed:	November 08, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240176279 A1	May. 30, 2024

Foreign Application Priority Data

JP	2022-188276	Nov. 25, 2022
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Publication Classification

Int. Cl.: G03G21/16 (20060101)

U.S. Cl.:

Field of Classification Search

CPC: G03G (21/1633); G03G (15/2053); G03G (21/168); G03G (15/1605); G03G (15/161)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6571073	12/2002	Suzuki	399/116	G03G 21/1619
2008/0219700	12/2007	Saiki	399/122	G03G 21/1842
2014/0174888	12/2013	Koshida	198/497	G03G 15/00
2018/0136603	12/2017	Watanabe	N/A	G03G 21/168
2021/0356888	12/2020	Chikugo	N/A	G03G 15/2053

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2000-098676	12/1999	JP	N/A

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present disclosure relates to an image forming apparatus including a belt unit.

2. Description of the Related Art

(2) There is a known image forming apparatus that includes an apparatus body and a belt unit attachable to and detachable from the apparatus body (see, for example, Japanese Patent Application Publication No. 2000-98676 (see FIG. 9)).

(3) However, in such conventional image forming apparatuses, if a belt unit is incorrectly attached to an irregular position different from a regular attachment position, there is a possibility that a user may not notice the incorrect attachment.

SUMMARY OF THE INVENTION

(4) The present disclosure has been made to solve the above problems, and an object of the present disclosure is to provide an image forming apparatus with a simple configuration that enables a user to easily recognize the incorrect attachment of a belt unit.

(5) An image forming apparatus of the present disclosure includes: an apparatus body, a fixing unit attachable to the apparatus body in a first direction, and a belt unit attachable in an attachment position within the apparatus body in a second direction intersecting the first direction. The belt unit has a restricting portion that restricts the attachment of the fixing unit when the belt unit is positioned in a first irregular position different from the attachment position.

(6) According to the present disclosure, the attachment of the fixing unit is restricted by the restricting portion when the belt unit is positioned in the first irregular position different from the

attachment position. Therefore, with a simple configuration, it becomes possible to enable the user to easily recognize incorrect attachment of the belt unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram illustrating a configuration of an image forming apparatus of an embodiment.
- (2) FIG. 2 is a diagram illustrating a configuration of a transfer unit of the embodiment.
- (3) FIG. 3 is a perspective view illustrating an appearance of the image forming apparatus of the embodiment.
- (4) FIG. 4 is a perspective view illustrating a state in which a top cover unit and a front cover in the image forming apparatus of the embodiment are opened.
- (5) FIG. 5 is a side cross-sectional view illustrating the state in which the top cover unit and the front cover in the image forming apparatus of the embodiment are opened.
- (6) FIG. 6 is a perspective view illustrating the transfer unit of the embodiment.
- (7) FIG. 7(A) is a perspective view illustrating the transfer unit of the embodiment, and FIG. 7(B) is an enlarged perspective view illustrating a part of the transfer unit.
- (8) FIG. 8(A) is a perspective view illustrating the transfer unit of the embodiment, and FIG. 8(B) is an enlarged perspective view illustrating a part of the transfer unit.
- (9) FIGS. 9(A) and 9(B) are diagrams of one of bearing portions of the transfer unit of the embodiment as viewed from $-X$ and $+X$ sides, respectively.
- (10) FIGS. 10(A) and 10(B) are diagrams of the other bearing portion of the transfer unit of the embodiment as viewed from $-X$ and $+X$ sides, respectively.
- (11) FIG. 11(A) is a perspective view illustrating a state in which the transfer unit is detached from a housing of the image forming apparatus of the embodiment, and FIG. 11(B) is an enlarged perspective view illustrating a part of the housing.
- (12) FIG. 12(A) is a perspective view illustrating a state in which the transfer unit is detached from the housing of the image forming apparatus of the embodiment, and FIG. 12(B) is an enlarged perspective view illustrating a part of the housing.
- (13) FIG. 13 is a perspective view illustrating an inlet opening of a fixing unit and its surroundings of the image forming apparatus of the embodiment.
- (14) FIGS. 14(A) and 14(B) are a perspective view and a side cross-sectional view illustrating an attachment operation of the transfer unit of the embodiment to the housing, respectively.
- (15) FIGS. 15(A) and 15(B) are side cross-sectional views illustrating a bearing portion and its surroundings when the transfer unit of the embodiment is attached in a regular attachment position.
- (16) FIG. 16 is a side cross-sectional view illustrating a state in which the transfer unit is incorrectly attached to the housing with the fixing unit attached thereto in the image forming apparatus of the embodiment.
- (17) FIGS. 17(A) and 17(B) are enlarged views of parts of the transfer unit illustrated in FIG. 16.
- (18) FIG. 18 is a side cross-sectional view illustrating a state in which the transfer unit is incorrectly attached to the housing to which the fixing unit is not attached in the image forming apparatus of the embodiment.
- (19) FIGS. 19(A) and 19(B) are diagrams illustrating a bearing portion of the transfer unit illustrated in FIG. 18.
- (20) FIGS. 20(A) and 20(B) are a perspective view and a side cross-sectional view illustrating an attachment operation of the fixing unit to the housing in the image forming apparatus of the embodiment, respectively.
- (21) FIGS. 21(A) and 21(B) are side cross-sectional views illustrating the action of restricting the

attachment of the fixing unit to the housing in the image forming apparatus of the embodiment.

(22) FIG. 22 is a diagram illustrating an inlet opening of the fixing unit and the transfer unit in the embodiment.

(23) FIGS. 23(A) and 23(B) are enlarged diagrams of the inlet opening of the fixing unit and the transfer unit in the embodiment.

DETAILED DESCRIPTION OF THE INVENTION

(24) <Configuration of Image Forming Apparatus 1>

(25) FIG. 1 is a diagram illustrating a configuration of an image forming apparatus 1 of a present embodiment. The image forming apparatus 1 forms images by using an electrophotographic method and is configured here as a printer that forms color images.

(26) As illustrated in FIG. 1, the image forming apparatus 1 includes a medium supply section 110, process units 10K, 10C, 10M, and 10Y, a transfer unit 20, a fixing unit 40, a medium ejection section 120, a housing 70, and a top cover unit 50. The medium supply section 110 supplies media P such as printing paper. The process units 10K, 10C, 10M, and 10Y form toner images (developer images). The transfer unit 20 transfers the toner image to the medium P. The fixing unit 40 fixes the toner image onto the medium P. The medium ejection section 120 ejects the medium P. The housing 70 accommodates therein these components described above. The top cover unit 50 is provided on the top of the housing 70.

(27) The medium supply section 110 has a paper feeding tray 111 that stores the media P, a pickup roller 112 disposed to contact the medium P stored in the paper feeding tray 111, a feed roller 113 disposed adjacent to the pickup roller 112, and a retard roller 114 disposed to face the feed roller 113.

(28) The paper feeding tray 111 stores a stack of the media P such as printing paper. The pickup roller 112 contacts the medium P in the paper feeding tray 111 and rotates to eject the medium P from the paper feeding tray 111. The feed roller 113 feeds the medium P ejected by the pickup roller 112 to a conveyance path R1. The retard roller 114 rotates in a direction opposite to a feeding direction taken by the feed roller 113 to apply conveyance resistance to the medium P to thereby prevents duplicate feeding.

(29) The medium supply section 110 also has a registration roller pair composed of a registration roller 115 and a pinch roller 116, and a conveyance roller pair 119 composed of a conveyance roller 117 and a pinch roller 118, along the conveyance path R1 for the medium P. The registration roller 115 starts rotating at predetermined timing after a leading end of the medium P contacts nip portions of the registration roller pair, and thereby it corrects the skew of the medium P and conveys the medium P. The conveyance roller 117 conveys the medium P from the registration roller pair to the transfer unit 20.

(30) The process units 10K, 10C, 10M, and 10Y (i.e., image forming units) are arranged in this order along a conveyance direction of the medium P. The process units 10K, 10C, 10M, and 10Y form black, cyan, magenta, and yellow toner images, respectively.

(31) Exposure heads 13K, 13C, 13M, and 13Y (i.e., exposure devices) are disposed so as to face respective photosensitive drums 11 (to be described later) of the process units 10K, 10C, 10M, and 10Y, respectively.

(32) The process units 10K, 10C, 10M, and 10Y are collectively referred to as “process units 10” when it is not necessary to distinguish them. The exposure heads 13K, 13C, 13M, and 13Y are collectively referred to as “exposure heads 13” when it is not necessary to distinguish them.

(33) The process unit 10 has the photosensitive drum 11 as an image carrier that carries the toner image, a charging roller 12 as a charging member, a developing roller 14 as a developer carrier, and a supply roller 15 as a supply member.

(34) The photosensitive drum 11 is a cylindrical member with a photosensitive layer (charge generation layer and charge transfer layer) provided on a surface of a conductive substrate, and is rotated by a drive motor clockwise as illustrated in FIG. 1.

(35) The charging roller **12** is disposed to contact the surface of the photosensitive drum **11** and rotates following the rotation of the photosensitive drum **11**. The charging roller **12** is applied with a charging voltage by a charging voltage power source, and uniformly charges the surface of the photosensitive drum **11**. An electrostatic latent image is formed on the uniformly charged surface of the photosensitive drum **11** by irradiation with light from an exposure head **13**.

(36) The exposure head **13** includes, for example, a light-emitting element array in which light-emitting elements such as LEDs (light-emitting diodes) are arranged, and a lens array that focuses the light emitted from the light-emitting elements onto the surface of the photosensitive drum **11**. The exposure head **13** causes the surface of the photosensitive drum **11** to be exposed to light, thereby forming the electrostatic latent image.

(37) The developing roller **14** is disposed to contact the surface of the photosensitive drum **11** and rotates in the direction (counterclockwise in the figure) opposite to that of the photosensitive drum **11**. The developing roller **14** is applied with a developing voltage by a developing voltage power source, and causes toner to adhere to the electrostatic latent image formed on the surface of the photosensitive drum **11**, thereby forming a toner image.

(38) The supply roller **15** is disposed to contact or face the surface of the developing roller **14** and rotates in the same direction (counterclockwise in the figure) as the developing roller **14**. The supply roller **15** is applied with a supply voltage a supply voltage power source, and supplies the toner to the developing roller **14**.

(39) The transfer unit **20** is disposed under the process units **10K**, **10C**, **10M**, and **10Y**. The transfer unit **20** is also referred to as a belt unit.

(40) FIG. 2 is a diagram illustrating the transfer unit **20**. The transfer unit **20** includes: transfer rollers **21** facing the respective photosensitive drums **11** of the process units **10K**, **10C**, **10M**, and **10Y**, an endless transfer belt **22** that passes between the photosensitive drum **11** and the transfer rollers **21**, and a drive roller **23** and a tension roller **24** around which the transfer belt **22** is stretched. The transfer belt **22** is also simply referred to as the belt.

(41) The transfer roller **21** is applied with a transfer voltage by a transfer voltage power source, and transfers the toner image formed on the photosensitive drum **11** to the medium P. The transfer belt **22** travels with the medium P adsorbed and held on its surface by electrostatic force. The drive roller **23** rotates by means of a belt motor, causing the transfer belt **22** to travel. The tension roller **24** is a driven roller and applies tension to the transfer belt **22**.

(42) Under the transfer belt **22**, a cleaning blade **202** is disposed in contact with an outer circumference of the transfer belt **22**. On an inner circumference of the transfer belt **22**, a backup roller **201** is disposed in a location where the transfer belt **22** is sandwiched between the cleaning blade **202** and the backup roller **201**. In an example illustrated in FIG. 2, the backup roller **201** and the cleaning blade **202** are disposed, for example, between the two transfer rollers **21** located at the most downstream side in the conveyance direction of the medium P.

(43) The cleaning blade **202** scrapes off the toner and the like adhering to the transfer belt **22**. A waste toner box **25** that stores the toner scraped off by the cleaning blade **202** is disposed under the transfer belt **22**.

(44) The waste toner box **25** extends in the conveyance direction of the medium P, for example, from a position corresponding to the second transfer roller **21** counted from the most downstream side to a position corresponding to the tension roller **24**. Within the waste toner box **25**, a conveyor/agitator (agitation bar) **203** is provided to spread the toner scraped off by the cleaning blade **202** throughout the waste toner box **25**.

(45) As illustrated in FIG. 1, the fixing unit **40** is disposed downstream of the transfer unit **20** in the conveyance direction of the medium P. The fixing unit **40** has, for example, a fixing belt **41**, a heater **43** disposed on the inner circumferential side of the fixing belt **41**, a pressure roller **42** pressed against the fixing belt **41**, and a unit housing **44** that accommodates therein these components.

(46) The fixing belt **41** is heated from its inner circumferential side by the heater **43**. The pressure roller **42** rotates with the driving force of a fixing motor, and the fixing belt **41** rotates following the pressure roller **42**. The fixing belt **41** and the pressure roller **42** apply heat and pressure, respectively, to the toner image transferred to the medium **P** to fix the toner image onto the medium **P**. Instead of the fixing belt **41** and heater **43**, a fixing roller with a built-in heat source may be used.

(47) The medium ejection section **120** is disposed downstream of the fixing unit **40** in the conveyance direction of the medium **P** and includes two pairs of rollers, i.e., ejection rollers **121** and **122**. The ejection rollers **121** and **122** convey the medium **P** fed from the fixing unit **40** along an ejection conveyance path **R2** and then eject the medium **P** to the outside of the image forming apparatus **1**. A stacker portion **123** for placing the media **P** ejected by the ejection rollers **121** and **122** is provided at the top of the top cover unit **50**.

(48) For double-sided printing, the image forming apparatus **1** is equipped with a re-conveyance section **130** that reverses the medium **P** with the toner image fixed on its surface and conveys the medium **P** back to the conveyance roller pair **119** described above. A switching guide **125** that guides the medium **P** fed from the fixing unit **40** to the medium ejection section **120** or re-conveyance section **130** is provided downstream of the fixing unit **40**.

(49) The re-conveyance section **130** has a return conveyance path **R3** that merges with the conveyance path **R1** on the upstream side of the conveyance roller pair **119**. Along the return conveyance path **R3**, conveyance rollers **131**, **132**, and **133** are provided. A pinch roller **134** is provided near an exit of the return conveyance path **R3** so as to contact the registration roller **115** from the opposite side to the pinch roller **116**.

(50) The conveyance rollers **131** to **133** convey the medium **P** along the return conveyance path **R3**. The roller pair of the registration roller **115** and the pinch roller **134** convey the medium **P** that has been conveyed along the return conveyance path **R3** to the conveyance roller pair **119**. In a case where the image forming apparatus **1** does not have the function of double-sided printing, the re-conveyance section **130** is not necessarily provided.

(51) The image forming apparatus **1** also has the top cover unit **50** as a cover member that is openable and closable and mounted at the top of the housing **70**, and a front cover **90** that is openable and closable and mounted onto a side portion of the housing **70**.

(52) The top cover unit **50** has toner cartridges **53K**, **53C**, **53M**, and **53Y** (see FIG. 4 to be described later) as developer storages that supply toner to the process units **10K**, **10C**, **10M**, and **10Y**, respectively. The toner cartridges **53K**, **53C**, **53M**, and **53Y** store black, cyan, magenta, and yellow toner, respectively. The top cover unit **50** is also provided with an operation panel **71** (FIG. 3) for performing an input operation to the image forming apparatus **1**.

(53) In FIG. 1, the direction of the rotation axis of the photosensitive drum **11** is defined as the **X** direction. The **X** direction is the width direction of the medium **P**. The axis direction of each roller in each unit described above is parallel to the **X** direction. The depth (front-rear) direction of the image forming apparatus **1** that is perpendicular to the **X** direction is defined as the **Y** direction. The **XY** plane here is the horizontal plane. The direction perpendicular to the **XY** plane, here the vertical direction, is defined as the **Z** direction. Specifically, the **Y** direction toward the side of the front cover **90** is defined as the **-Y** direction (rearward), whereas the direction toward its opposite side is the **+Y** direction (forward).

(54) In FIG. 1, the process units **10K**, **10C**, **10M**, and **10Y** are arranged in a direction inclined by about 10 degrees relative to the **Y** direction. The arrangement direction of the transfer rollers **21** of the transfer unit **20** is parallel to the arrangement direction of the process units **10K**, **10C**, **10M**, and **10Y**. However, these arrangement directions do not necessarily have to be inclined relative to the **Y** direction, and may be parallel to the **Y** direction.

(55) Of the components of the image forming apparatus **1**, the fixing unit **40** is configured to be attachable to and detachable from the housing **70** in the directions indicated by arrows **A1** and **A2**, respectively. The direction indicated by the arrow **A1** is the attachment direction (first direction),

whereas the direction indicated by the arrow A2 is the detachment direction. These directions are collectively referred to as the attachment/detachment direction (A direction) of the fixing unit **40**.

(56) The attachment/detachment direction (A direction) of the fixing unit **40** is here inclined by about 10 degrees relative to the Z direction. That is, the attachment direction of the fixing unit **40** is substantially $-Z$ direction, and the detachment direction of the fixing unit **40** is substantially $+Z$ direction. However, the attachment/detachment direction of the fixing unit **40** is not limited to this direction.

(57) The transfer unit **20** is configured to be attachable to and detachable from the housing **70** in the directions indicated by arrows B1 and B2, respectively. The direction indicated by the arrow B1 is defined as the attachment direction (second direction), whereas the direction indicated by the arrow B2 is the detachment direction. These directions are collectively referred to as the attachment/detachment direction (B direction) of the transfer unit **20**.

(58) The attachment/detachment direction (B direction) of the transfer unit **20** is here inclined by about 10 degrees relative to the Y direction. That is, the attachment direction of the transfer unit **20** is substantially $+Y$ direction, and the detachment direction of the transfer unit **20** is substantially $-Y$ direction. However, the attachment/detachment direction of the transfer unit **20** is not limited to this direction.

(59) The attachment/detachment direction (A direction) of the fixing unit **40** and the attachment/detachment direction (B direction) of the transfer unit **20** are perpendicular to each other in an example illustrated in FIG. 1. However, these directions do not necessarily have to be perpendicular to each other, and they only need to intersect each other.

(60) The portion of the image forming apparatus **1**, excluding the transfer unit **20**, fixing unit **40**, top cover unit **50**, and front cover **90**, is referred to as an apparatus body **100**. The apparatus body **100** includes the housing **70**, the medium supply section **110**, the process units **10K**, **10C**, **10M**, and **10Y**, the medium ejection section **120**, and the like.

(61) FIG. 3 is a perspective view illustrating an appearance of the image forming apparatus **1**. The top cover unit **50** is provided on the housing **70** so as to be rotatable about a rotation axis C1 in the X direction. The rotation axis C1 is provided at an end (front end) of the housing **70** in the $+Y$ direction and at an end (top end) thereof in the $+Z$ direction.

(62) The front cover **90** is provided on the housing **70** so as to be rotatable about a rotation axis C2 in the X direction. The rotation axis C2 is provided at an end (rear end) of the housing **70** in the $-Y$ direction and at an end (bottom end) thereof in the $-Z$ direction.

(63) FIG. 4 is a perspective view illustrating a state in which the top cover unit **50** and the front cover **90** in the image forming apparatus **1** both are rotated to respective open positions. The top cover unit **50** is rotatable between a closed position illustrated in FIG. 3 and the open position shown in FIG. 4. The front cover **90** is rotatable between a closed position illustrated in FIG. 3 and the open position shown in FIG. 4.

(64) Of the components of the image forming apparatus **1**, the toner cartridges **53K**, **53C**, **53M**, and **53Y** and the exposure heads **13K**, **13C**, **13M**, and **13Y** (FIG. 1) are held by the top cover unit **50**.

(65) The top cover unit **50** has a unit frame **52** that is rotatable about the rotation axis C1 and a cover **51** that covers the top of the unit frame **52**.

(66) The unit frame **52** holds the toner cartridges **53K**, **53C**, **53M**, and **53Y** arranged in the X direction such that each of these toner cartridges **53K**, **53C**, **53M**, and **53Y** can be pulled out therefrom. The cover **51** is provided with an opening **56** (FIG. 3) for attaching and detaching the fixing unit **40** to and from the housing **70**.

(67) The unit frame **52** is provided with a toner conveyance path **200** (FIG. 1) for conveying toner from the toner cartridges **53K**, **53C**, **53M**, and **53Y** to the process units **10K**, **10C**, **10M**, and **10Y**.

(68) The image forming apparatus **1** also has a basket unit **60** that holds the process units **10K**, **10C**, **10M**, and **10Y**. The basket unit **60** has a rotatable frame **61** that is rotatable about the rotation axis C2 described above, and a basket **62** held by the rotatable frame **61**. The basket **62** holds the

process units **10K**, **10C**, **10M**, and **10Y**.

(69) The top cover unit **50** and the basket unit **60** are rotatable in a state where the top cover unit **50** and the basket unit **60** are connected to each other. However, it is also possible that only the top cover unit **50** is rotatable alone. The top cover unit **50** is provided with a first operation lever **54** and a second operation lever **55**. The first operation lever **54** is operated when the top cover unit **50** is opened or closed. The second operation lever **55** is operated when the connection state between the top cover unit **50** and the basket unit **60** is switched.

(70) The first operation lever **54** is interlocked with a locking portion **58** provided on the unit frame **52**. When the top cover unit **50** is located in the closed position, the locking portion **58** is engaged with a hole **78** formed in an end surface portion **75** of the housing **70**. That is, the top cover unit **50** is locked in the closed position with respect to the housing **70**. When the first operation lever **54** is pulled out, the locking portion **58** is moved out of the hole **78** in the housing **70** to unlock the top cover unit **50**.

(71) The second operation lever **55** is operated to connect and disconnect the top cover unit **50** and the basket unit **60**. In a state where the second operation lever **55** is pushed in, the top cover unit **50** and the basket unit **60** are connected together. On the other hand, in a state where the second operation lever **55** is pulled out, the top cover unit **50** and the basket unit **60** are disconnected.

(72) The second operation lever **55** is also interlocked with a shutter (not shown) that opens and closes the toner conveyance path **200** through which toner is supplied from the toner cartridges **53K**, **53C**, **53M**, and **53Y** to the process units **10K**, **10C**, **10M**, and **10Y**.

(73) In a state where the second operation lever **55** is pushed in, the toner can be supplied from the toner cartridges **53K**, **53C**, **53M**, and **53Y** to the process units **10K**, **10C**, **10M**, and **10Y** via the toner conveyance path **200**. In a state where the second operation lever **55** is pulled out, the supply of the toner via the toner conveyance path **200** is blocked.

(74) In the state illustrated in FIGS. **4** and **5**, the second operation lever **55** is pushed in, and the top cover unit **50** and the basket unit **60** are rotated to the open position in a state where they are connected together.

(75) By the rotation of the top cover unit **50** and basket unit **60** to the open position, the top of the transfer unit **20** is opened, and the transfer unit **20** can be detached from the housing **70**.

(76) A description of the specific configuration of the top cover unit **50** and the basket unit **60** will be omitted. The image forming apparatus **1** is not limited to that having the top cover unit **50** and basket unit **60** as described above. For example, the image forming apparatus **1** may be configured such that toner cartridges are attachable to and detachable from the process units **10K**, **10C**, **10M**, and **10Y**.

(77) <Configuration of Transfer Unit **20**>

(78) FIG. **6** is a perspective view illustrating the transfer unit **20**. The transfer unit **20** has a belt frame **26** that supports the transfer belt **22** and four transfer rollers **21** (FIG. **2**). The extending direction of the belt frame **26** is substantially the Y direction in a state where the transfer unit **20** is attached to the housing **70**.

(79) On the lower side of the belt frame **26**, the waste toner box **25** described above is provided. Meanwhile, on the upper side of the belt frame **26**, a gripping portion **30** to be gripped by a user upon attachment or detachment of the transfer unit **20** is provided. The gripping portion **30** is located at the end (rear end) of the belt frame **26** substantially in the -Y direction and extends in the X direction.

(80) FIG. **7(A)** is a perspective view of the transfer unit **20** as viewed from the left rear side. FIG. **7(B)** is an enlarged perspective view of a part of the transfer unit **20** illustrated in FIG. **7(A)**. FIG. **8(A)** is a perspective view of the transfer unit **20** as viewed from the right rear side. FIG. **8(B)** is an enlarged perspective view of a part of the transfer unit **20** illustrated in FIG. **8(A)**.

(81) As illustrated in FIG. **7(A)**, on the -X side of the belt frame **26**, a U-shaped groove **28L** opened substantially in the -Z direction is formed at the center region substantially in the Y

direction. The U-shaped groove **28L** serves as a positioning portion that positions the transfer unit **20** by engaging a post **84L** (FIG. **11(B)**) provided inside the housing **70**.

(82) As illustrated in FIG. **8(A)**, a U-shaped groove **28R** opened substantially in the $-Z$ direction is formed on the $+X$ side of the belt frame **26**. The U-shaped groove **28R** by engages a post **84R** (FIG. **12(B)**) provided inside the housing **70** and serves as a positioning portion that positions the transfer unit **20**.

(83) As illustrated in FIGS. **7(A)** and **8(A)**, the transfer unit **20** has bearing portions **31L** and **31R** on both sides of the transfer belt **22** in the X direction. The bearing portions **31L** and **31R** are located substantially in the $+Y$ side (forward) of the transfer belt **22**.

(84) The bearing portion **31L** is mounted on the $-X$ side of the belt frame **26** and rotatably supports a shaft **23a** of the drive roller **23** as illustrated in an enlarged view of FIG. **7(B)**. The bearing portion **31L** is formed of resin, such as polyacetal (POM), for example.

(85) The bearing portion **31R** is mounted on the $+X$ side of the belt frame **26** and rotatably supports the shaft **23a** of the drive roller **23** as illustrated in an enlarged view of FIG. **8(B)**. The bearing portion **31R** is formed of resin, such as POM, for example.

(86) At the end of the drive roller **23** in the $+X$ direction, a gear **27** is fixed to transmit rotation from the belt motor to the drive roller **23**.

(87) FIG. **9(A)** is a diagram of the bearing portion **31L** as viewed from the $-X$ side. FIG. **9(B)** is a diagram of the bearing portion **31L** as viewed from the $+X$ side. As illustrated in FIGS. **9(A)** and **9(B)**, the bearing portion **31L** has a fitting portion **31c** that supports the shaft **23a** (FIG. **7(B)**) of the drive roller **23**. The fitting portion **31c** is formed substantially in an annular shape about a rotation axis **C3**.

(88) The fitting portion **31c** has a support hole **31h** into which the shaft **23a** is inserted. The support hole **31h** is a circular hole about the rotation axis **C3** of the shaft **23a**. The fitting portion **31c** also engages a positioning groove **82L** (FIG. **11(B)**) serving as a first engagement portion to be described later when the transfer unit **20** is attached. Therefore, the fitting portion **31c** is also referred to as a second engagement portion.

(89) The bearing portion **31L** also has an arm **31f** located on substantially the $+Y$ side of the fitting portion **31c**, a protrusion **31a** as a restricting portion protruding from the arm **31f** substantially in the $+Y$ direction, and a guide portion **31b** extending in an arc shape substantially in the $-Z$ direction (downward) from the arm **31f**.

(90) The protrusion **31a** is a contact portion that contacts the unit housing **44** of the fixing unit **40** when the transfer unit **20** is incorrectly attached. The protrusion **31a** is a plate-shaped member having a thickness in the X direction and is elastically deformable in the X direction.

(91) The guide portion **31b** is formed in an arc shape about the rotation axis **C3**. The guide portion **31b** is a sliding portion that slides along a guide rail **81L** (FIG. **11(B)**) to be described later when the transfer unit **20** is attached.

(92) FIG. **10(A)** is a diagram of the bearing portion **31R** as viewed from the $-X$ side. FIG. **10(B)** is a diagram of the bearing portion **31R** as viewed from the $+X$ side. As illustrated in FIGS. **10(A)** and **10(B)**, the bearing portion **31R** has a shape symmetrical to the bearing portion **31L** relative to the YZ plane. That is, the bearing portion **31R** has the protrusion **31a** as the restricting portion, the guide portion **31b**, the fitting portion **31c** as the second engagement portion, and the arm **31f**.

(93) The protrusion **31a** is a contact portion that contacts the fixing unit **40** when the transfer unit **20** is incorrectly attached. The guide portion **31b** is a sliding portion that slides along a guide rail **81R** (FIG. **12(B)**) when the transfer unit **20** is attached. The fitting portion **31c** has the support hole **31h** and thereby supports the shaft **23a** (FIG. **8(B)**) of the drive roller **23**. The fitting portion **31c** also engages a positioning groove **82R** (FIG. **12(B)**) to be described later when the transfer unit **20** is attached.

(94) As illustrated in FIGS. **7(A)** and **8(A)**, ribs **25e** are formed near both ends of the waste toner box **25** in the X direction at the end surface thereof in the $-Y$ direction. The ribs **25e** contact a roller

frame 85 (FIG. 17(B)) in the housing 70 when the transfer unit 20 is incorrectly attached.

(95) FIG. 11(A) is a perspective diagram illustrating a state in which the transfer unit 20 is detached from the housing 70 of the image forming apparatus 1. FIG. 11(B) is an enlarged perspective diagram illustrating a part of the housing 70 illustrated in FIG. 11(A).

(96) As illustrated in FIG. 11(A), an accommodation area 72 for accommodating the transfer unit 20 is formed inside the housing 70. The accommodation area 72 is formed between side walls 74 on both sides of the housing 70 in the X direction.

(97) As illustrated in FIG. 11(B), the guide rail 81L is provided at an end of the accommodation area 72 in the -X direction. The guide rail 81L extends substantially in the Y direction, i.e., in the second direction described above. The guide rail 81L is a guide member that guides the guide portion 31b (FIG. 7(B)) of the bearing portion 31L of the transfer unit 20.

(98) The guide rail 81L has a slope 811 and a top portion 812 at a region substantially in the +Y direction (forward) from the central region of the guide rail 81L. The slope 811 is inclined upward, and the top portion 812 leads to a front end of the slope 811.

(99) The positioning groove 82L (i.e., a first engagement portion) is formed adjacent to the guide rail 81L. The positioning groove 82L engages the fitting portion 31c (FIG. 9(B)) of the bearing portion 31L of the transfer unit 20. The positioning groove 82L is, for example, a U-shaped groove opened substantially in the -Y direction.

(100) The positioning groove 82L is formed at a position where the fitting portion 31c (FIG. 9(B)) is fitted to the positioning groove 82L when the guide portion 31b (FIG. 9(B)) of the bearing portion 31L reaches the top portion 812 of the guide rail 81L. The positioning groove 82L is formed, for example, in a plate 80L disposed adjacent to the guide rail 81L in the -X direction.

(101) The post 84L that engages the U-shaped groove 28L (FIG. 7(A)) of the transfer unit 20 is provided inside the housing 70. The post 84L is, for example, a shaft member whose axial direction is the X direction. The engagement between the post 84L and the U-shaped groove 28L positions the transfer unit 20 substantially in the Y direction.

(102) FIG. 12(A) is a perspective diagram illustrating a state in which the transfer unit 20 is detached from the housing 70 of the image forming apparatus 1 as viewed from the opposite side to FIG. 11(A). FIG. 12(B) is an enlarged perspective diagram illustrating a part of the housing 70 illustrated in FIG. 12(A).

(103) As illustrated in FIG. 12(B), the guide rail 81R is provided at an end of the accommodation area 72 in the +X direction. The guide rail 81R extends substantially in the Y direction, i.e., in the second direction described above. The guide rail 81R is a guide member that guides the guide portion 31b (FIG. 8(B)) of the bearing portion 31R of the transfer unit 20. The guide rail 81R has a slope 811 and a top portion 812, as in the guide rail 81L described above.

(104) The positioning groove 82R (as the first engagement portion) is formed adjacent to the guide rail 81R. The positioning groove 82R engages the fitting portion 31c (FIG. 10(A)) of the bearing portion 31R of the transfer unit 20. The positioning groove 82R is, for example, a U-shaped groove opened substantially in the -Y direction.

(105) The positioning groove 82R is formed at a position where the fitting portion 31c (FIG. 10(B)) is fitted to the positioning groove 82R when the guide portion 31b (FIG. 10(B)) of the bearing portion 31R reaches the top portion 812 of the guide rail 81R. The positioning groove 82R is formed, for example, in a plate 80R disposed adjacent to the guide rail 81R in the -X direction.

(106) The post 84R that engages the U-shaped groove 28R (FIG. 8(A)) of the transfer unit 20 is provided inside the housing 70. The post 84R is, for example, a shaft member whose axial direction is the X direction. The engagement between the post 84R and the U-shaped groove 28R positions the transfer unit 20 in the X direction and substantially in the Y direction.

(107) FIG. 13 is a perspective view illustrating a part of the fixing unit 40. An inlet opening 45 is formed in a wall portion 48 of the unit housing 44 in the fixing unit 40. The wall portion 48 faces the transfer unit 20. The inlet opening 45 is an opening for introducing the medium P conveyed by

the transfer belt **22** of the transfer unit **20** into the unit housing **44**.

(108) The inlet opening **45** has a rectangular shape that is elongated in the X direction. The inlet opening **45** has a restriction surface **45a** as a second contact portion which is an end of the inlet opening **45** substantially in the +Z direction, and a pair of end surfaces **45b** located at both ends of the inlet opening **45** in the X direction. The width of the inlet opening **45** in the X direction is wider than the width of the transfer belt **22**.

(109) Inclined surfaces **46L** and **46R** are provided on both sides of the inlet opening **45** in the X direction. The inclined surfaces **46L** and **46R** are inclined so as to be positioned in substantially the +Y direction along substantially the -Z direction (see FIGS. **15(A)** and **15(B)** to be described later).

(110) Retracted surfaces **47L** and **47R** are formed to be continuous to lower ends (ends substantially in the -Z direction) of the inclined surfaces **46L** and **46R**, respectively. The retracted surfaces **47L** and **47R** are parallel to the wall portion **48** of the unit housing **44** and are retracted substantially in the +Y direction from the wall portion **48**.

(111) <Operation of Image Forming Apparatus>

(112) Next, an image forming operation (printing operation) of the image forming apparatus **1** will be described with reference to FIG. **1**. A controller of the image forming apparatus **1** starts the image forming operation when the controller receives a printing command and print data from a host device.

(113) First, the pickup roller **112** ejects the medium P stored in the paper feeding tray **111**, and the feed roller **113** feeds the ejected medium P to the conveyance path R1. The retard roller **114** applies conveyance resistance to the medium P, thereby preventing duplicate feeding. The registration roller **115** and the conveyance roller **117** rotate to convey the medium P fed into the conveyance path R1 to the transfer unit **20**.

(114) In the transfer unit **20**, the drive roller **23** rotates to cause the transfer belt **22** to travel. The transfer belt **22** adsorbs and holds the medium P and conveys the medium P. The medium P passes through the process units **10K**, **10C**, **10M**, and **10Y** in this order.

(115) In each process unit **10**, a toner image of each color is formed. That is, the photosensitive drum **11** rotates, and the charging rollers **12**, the developing rollers **14**, and the supply rollers **15** rotate along with the rotation of the photosensitive drum **11**. The charging roller **12** uniformly charges the surface of the photosensitive drum **11**. The exposure head **13** exposes the surface of the photosensitive drum **11** with light based on image data for each color to form the electrostatic latent image.

(116) The electrostatic latent image formed on the surface of the photosensitive drum **11** is developed with the toner adhering to the developing roller **14** to form a toner image on the surface of the photosensitive drum **11**. The toner image on the photosensitive drum **11** is transferred to the medium P on the transfer belt **22** by the transfer voltage applied to the transfer roller **21**.

(117) The toner images of the respective colors formed by the process units **10K**, **10C**, **10M**, and **10Y** are sequentially transferred to the medium P and superimposed on each other. The medium P on which the toner images of the respective colors have been transferred is further conveyed by the transfer belt **22** and introduced into the fixing unit **40** via the inlet opening **45**.

(118) In the fixing unit **40**, heat and pressure are applied to the medium P by the fixing belt **41** heated by the heater **43** and also by the pressure roller **42**, respectively, whereby the toner images are fixed on the medium P.

(119) The medium P with the toner images of respective colors fixed thereon is conveyed along the ejection conveyance path R2 by the ejection rollers **121** and **122** of the medium ejection section **120**, and is ejected to the outside of the image forming apparatus **1**. The ejected media P is placed on the stacker portion **123**. With this, the image forming operation on the medium P is completed.

(120) In the case of double-sided printing, the ejection rollers **121** and **122** convey the medium P by a predetermined amount and reverse the medium P when a rear end of the medium P passes through the switching guide **125**. Furthermore, the switching guide **125** operates to feed the

medium P to the re-conveyance section **130**. In the return conveyance path **R3**, the medium P is conveyed by the conveyance rollers **131** to **133** and further passes through between the registration roller **115** and the pinch roller **134** to reach the conveyance path **R1**. Thereafter, the medium P is conveyed by the conveyance roller pair **119** to the transfer unit **20**, where the toner image is formed on the back surface of the medium P.

(121) When the above image forming operation is performed, the first operation lever **54** (FIG. **4**) is located at the locking position to lock the top cover unit **50** in the housing **70**. The second operation lever **55** is located at the connecting position to connect the top cover unit **50** to the basket unit **60**.

(122) On the other hand, when a jammed medium P is to be detached from the image forming apparatus **1** or when the transfer unit **20** is to be replaced, the top cover unit **50** and the basket unit **60** are opened in a state where the top cover unit **50** and the basket unit **60** are connected together as illustrated in FIG. **4**.

(123) The user first opens the front cover **90** and pulls out the first operation lever **54** to unlock the top cover unit **50**. Then, the user rotates the top cover unit **50** to the open position without pulling out the second operation lever **55**. This allows the top cover unit **50** and the basket unit **60** to be rotated to the open position in a state where the top cover unit **50** and the basket unit **60** are connected together, as illustrated in FIGS. **4** and **5**, thus enabling the replacement of the transfer unit **20**.

(124) If the second operation lever **55** is pulled out before the top cover unit **50** is rotated to the open position, the connection between the top cover unit **50** and the basket unit **60** is released, thus enabling the replacement of the process units **10K**, **10C**, **10M**, and **10Y**. A description of this will be omitted.

(125) <Attachment Operation of Transfer Unit **20**>

(126) Next, an operation of attaching the transfer unit **20** in the housing **70** of the image forming apparatus **1** will be described. FIGS. **14(A)** and **14(B)** are a perspective view and a side view illustrating an attachment operation of the transfer unit **20** in the housing **70** of the image forming apparatus **1**, respectively.

(127) As illustrated in FIG. **14(A)**, the user grasps the gripping portion **30** of the transfer unit **20** and inserts the transfer unit **20** into the housing **70**. At this time, as illustrated in FIG. **14(B)**, the user inserts the transfer unit **20** into the housing **70** while the guide portions **31b** of the bearing portions **31L** and **31R** slide along the guide rails **81L** and **81R** of the housing **70**, respectively.

(128) FIGS. **15(A)** and **15(B)** are cross-sectional views at different positions illustrating the bearing portion **31R** and its surroundings when the transfer unit **20** is attached in the attachment position. While FIGS. **15(A)** and **15(B)** illustrate the bearing portion **31R**, the action of the bearing portion **31L** is the same as that of the bearing portion **31R**.

(129) As illustrated in FIG. **15(B)**, when the transfer unit **20** is attached in the attachment position, the fitting portions **31c** of the bearing portions **31L** and **31R** fit into the positioning grooves **82L** and **82R** of the housing **70** substantially in the +Y direction (second direction), respectively.

(130) At this time, as illustrated in FIG. **15(A)**, the protrusions **31a** of the bearing portions **31L** and **31R** face the retracted surfaces **47L** and **47R** of the unit housing **44** of the fixing unit **40**, respectively. In this state, the protrusions **31a** do not contact any of the wall portion **48**, the inclined surfaces **46L** and **46R**, and the retracted surfaces **47L** and **47R** of the unit housing **44** of the fixing unit **40**.

(131) In a state where the transfer unit **20** is attached in the attachment position, the waste toner box **25** (FIG. **14(B)**) of the transfer unit **20** is placed on the guide rails **81L** and **81R** of the housing **70**. The U-shaped grooves **28L** and **28R** (FIG. **7(A)**, FIG. **8(A)**) of the transfer unit **20** are engaged with the posts **84L** and **84R** (FIG. **11(B)**, FIG. **12(B)**) of the housing **70**, respectively, to position the transfer unit **20** in the X, Y and Z directions. The position of the transfer unit **20** at this time is defined as the attachment position (or regular attachment position).

(132) After the transfer unit **20** is attached in the attachment position, when the user rotates the top cover unit **50** to the closed position, the top cover unit **50** and the basket unit **60** rotate together to the closed position. When the top cover unit **50** and the basket unit **60** reach the closed position together, the process units **10K**, **10C**, **10M**, and **10Y** of the basket unit **60** contact the surface of the transfer belt **22** of the transfer unit **20**.

(133) In this state, when the user rotates the top cover unit **50** to the closed position, the locking portion **58** illustrated in FIG. **4** engages the hole **78**, whereby the top cover unit **50** is locked to the housing **70**. Furthermore, when the user rotates the front cover **90** to the closed position and locks the front cover **90**, the image forming apparatus **1** is brought into a state where the image forming apparatus **1** is capable of start printing.

(134) <Incorrect Attachment of the Transfer Unit **20** (when the Fixing Unit **40** is Attached)>

(135) Next, the incorrect attachment of the transfer unit **20** to the housing **70** of the image forming apparatus **1** will be described. First, a description will be given on a case in which the transfer unit **20** is incorrectly attached to the housing **70** of the image forming apparatus **1** in a state where the fixing unit **40** is attached to the housing **70**.

(136) FIG. **16** is a side cross-sectional view illustrating a state in which the transfer unit **20** is incorrectly attached to the housing **70** of the image forming apparatus **1**. As illustrated in FIG. **16**, the user may insert the transfer unit **20** into the housing **70** while lifting the transfer unit **20** without causing the transfer unit **20** to slide along the guide rails **81L** and **81R**.

(137) FIG. **17(A)** is a cross-sectional view illustrating the bearing portion **31R** and its surroundings in the transfer unit **20** incorrectly attached. FIG. **17(B)** is a cross-sectional view illustrating the waste toner box **25** and its surroundings in the transfer unit **20** incorrectly attached. While FIG. **17(A)** illustrates the bearing portion **31R**, the action of the bearing portion **31L** is the same as that of the bearing portion **31R**.

(138) When the transfer unit **20** is inserted while being lifted, each of the protrusions **31a** of the bearing portions **31L** and **31R** of the transfer unit **20** contacts the wall portion **48** of the unit housing **44** of the fixing unit **40**. The movement of the transfer unit **20** substantially in the +Y direction (second direction) is restricted by the contact between the wall portion **48** and each protrusion **31a** of the bearing portions **31L** and **31R**.

(139) Thus, as illustrated in FIG. **17(B)**, the ribs **25e** of the waste toner box **25** of the transfer unit **20** contact the roller frame **85** provided in the housing **70**. Thus, the transfer unit **20** cannot be descended to the position where the transfer unit **20** contacts the guide rails **81L** and **81R**.

(140) When the user intends to rotate the top cover unit **50** to the closed position in this state, the process units **10K**, **10C**, **10M**, and **10Y** of the basket unit **60** connected to the top cover unit **50** contact the transfer belt **22**. Thus, the top cover unit **50** cannot be rotated to the closed position.

(141) That is, as a result of the restriction of the movement of the transfer unit **20** in the +Y direction, the closing of the top cover unit **50** is also restricted. The user can recognize that the transfer unit **20** has been attached incorrectly since the top cover unit **50** cannot be closed.

(142) As illustrated in FIGS. **17(A)** and **17(B)**, the position of the transfer unit **20** when the protrusions **31a** of the bearing portions **31L** and **31R** contact the wall portion **48** of the fixing unit **40** and the waste toner box **25** contacts the roller frame **85** is referred to as a second irregular position.

(143) <Incorrect Attachment of the Transfer Unit **20** (in a State where the Fixing Unit **40** is not Attached)>

(144) Next, a description will be given on a case in which the transfer unit **20** is incorrectly attached to the housing **70** of the image forming apparatus **1** in a state where the fixing unit **40** is not attached to the housing **70**.

(145) FIG. **18** is a side cross-sectional view illustrating a state in which the transfer unit **20** is incorrectly attached to the housing **70** of the image forming apparatus **1** in the state where the fixing unit **40** is not attached to the housing **70**. As illustrated in FIG. **18**, the housing **70** of the

image forming apparatus **1** has an accommodation area **73** in which the fixing unit **40** is accommodated and which is adjacent in the +Y direction to the accommodation area **72** of the transfer unit **20**. The fixing unit **40** is attached in the accommodation area **73** substantially in the -Z direction (first direction) along a guide rail **86** provided in the housing **70**.

(146) FIGS. **19(A)** and **19(B)** are cross-sectional views at different cross-sections illustrating the bearing portion **31R** and its surroundings of the transfer unit **20** incorrectly attached in the state where the fixing unit **40** is not attached to the housing **70**. While FIGS. **19(A)** and **19(B)** illustrate the bearing portion **31R**, the action of the bearing portion **31L** is the same as that of the bearing portion **31R**.

(147) As illustrated in FIG. **19(A)**, when the user inserts the transfer unit **20** into the housing **70** while lifting the transfer unit **20** without causing the transfer unit **20** to slide along the guide rails **81L** and **81R**, there is nothing that contacts the protrusions **31a** of the bearing portions **31L** and **31R** because the fixing unit **40** is not attached. That is, the movement of the transfer unit **20** substantially in the +Y direction cannot be restricted.

(148) Thus, as illustrated in FIG. **19(B)**, the fitting portions **31c** of the bearing portions **31L** and **31R** ride onto shelves **83** of the positioning grooves **82L** and **82R** in the +Z direction.

(149) In this case, the waste toner box **25** of the transfer unit **20** does not contact the roller frame **85** (FIG. **17(B)**). As a result, the transfer unit **20** is brought into a state where the bearing portions **31L** and **31R** ride onto the shelf **83** and the waste toner box **25** is placed on the guide rails **81L** and **81R**. The position of the transfer unit **20** at this time is referred to as a first irregular position.

(150) FIGS. **20(A)** and **20(B)** are a perspective view and a side cross-sectional view illustrating an attachment operation of the fixing unit **40** in the housing **70** of the image forming apparatus **1**, respectively. As illustrated in FIGS. **20(A)** and **20(B)**, the fixing unit **40** is attached in the accommodation area **73** within the housing **70** in the first direction (substantially in the -Z direction) via an opening **56** formed in the top cover unit **50**.

(151) FIGS. **21(A)** and **21(B)** are cross-sectional views illustrating the attachment operation of the fixing unit **40** in a state in which the transfer unit **20** is incorrectly attached to the housing **70** of the image forming apparatus **1**. While FIGS. **21(A)** and **21(B)** illustrate the bearing portion **31R**, the action of the bearing portion **31L** is also the same as that of the bearing portion **31R**.

(152) As illustrated in FIG. **21(A)**, when the fixing unit **40** is inserted into the housing **70** substantially in the -Z direction, the inclined surfaces **46L** and **46R** of the unit housing **44** contact the protrusions **31a** of the bearing portions **31L** and **31R** of the transfer unit **20**, respectively. Thus, the user perceives a light impact due to the contact between the inclined surfaces **46L** and **46R** and the protrusions **31a** of the bearing portions **31L** and **31R**.

(153) When the fixing unit **40** is further inserted into the housing **70** substantially in the -Z direction, the protrusions **31a** of the bearing portions **31L** and **31R** are elastically deformed in the X direction. Thus, as illustrated in FIG. **21(B)**, the protrusions **31a** of the bearing portions **31L** and **31R** enter the interior of the inlet opening **45** through the inclined surfaces **46L** and **46R**, respectively.

(154) When the fixing unit **40** is further inserted into the housing **70** substantially in the -Z direction, the protrusions **31a** of the bearing portions **31L** and **31R** contact the upper end (i.e., restriction surface **45a**) of the inlet opening **45**. The movement of the fixing unit **40** substantially in the -Z direction is restricted by the contact between the protrusion **31a** and the restriction surface **45a**.

(155) Thus, the user can recognize that the transfer unit **20** has been attached incorrectly since the fixing unit **40** can be inserted only halfway.

(156) FIG. **22** is a diagram of the inlet opening **45** of the fixing unit **40** and a part of the transfer unit **20** facing the inlet opening **45** as viewed in a direction indicated by arrow E in FIG. **13**. FIG. **23(A)** is a diagram illustrating an end region of the inlet opening **45** in the -X direction and a part of the transfer unit **20** facing this end region. FIG. **23(B)** is a diagram illustrating an end region of

the inlet opening **45** in the +X direction and a part of the transfer unit **20** facing this end region.

(157) As illustrated in FIG. **22**, a width **W1** of the inlet opening **45** of the fixing unit **40** in the X direction is wider than a width **W2** of the transfer belt **22** in the X direction. The inclined surfaces **46L** and **46R** are formed on both sides of the inlet opening **45** in the X direction in the unit housing **44**.

(158) As illustrated in FIG. **23(A)**, the protrusion **31a** of the bearing portion **31L** of the transfer unit **20** is disposed to straddle a boundary portion between the inlet opening **45** and the inclined surface **46L**. As illustrated in FIG. **23(B)**, the protrusion **31a** of the bearing portion **31R** of the transfer unit **20** is disposed to straddle a boundary portion between the inlet opening **45** and the inclined surface **46R**.

(159) Thus, when the protrusions **31a** of the bearing portions **31L** and **31R** contact the inclined surfaces **46L** and **46R** of the unit housing **44**, respectively, the protrusions **31a** elastically deform in the X direction to enter the inlet opening **45**, and can be brought into a state of contacting the restriction surface **45a**.

(160) That is, when the user inserts the fixing unit **40** in the state where the transfer unit **20** is incorrectly attached, the incorrect attachment of the transfer unit **20** can be recognized by the light impact caused by the contact between the protrusions **31a** of the bearing portions **31L** and **31R** and the unit housing **44**, and also by the restriction of the attachment of the fixing unit **40** due to the contact between the protrusions **31a** and the restriction surface **45a**.

(161) In particular, as illustrated in FIGS. **21(A)**, the inclined surfaces **46L** and **46R** of the unit housing **44** are inclined toward the restriction surface **45a** (i.e., inclined so as to guide the protrusion **31a** toward the restriction surface **45a**). Thus, the impact from the contact between the protrusions **31a** and the inclined surfaces **46L** and **46R** can be mitigated, and the protrusion **31a** can be smoothly brought into a state of contacting the restriction surface **45a**.

(162) The contact between the protrusion **31a** and the restriction surface **45a** does not allow the fixing unit **40** to be accommodated in the accommodation area **73** and causes the fixing unit **40** to protrude upward from the image forming apparatus **1**, thereby enabling the user to easily recognize the incorrect attachment.

(163) Thus, the user can recognize that the transfer unit **20** has been attached incorrectly without causing any damage to components such as the bearing portions **31L** and **31R**.

(164) <Effects of Embodiment>

(165) As described above, the image forming apparatus **1** of the present embodiment includes the apparatus body **100**, the fixing unit **40** attachable to the apparatus body **100** in the first direction (substantially in the -Z direction), and the transfer unit (belt unit) **20** attachable in the attachment position within the apparatus body **100** in the second direction (substantially in the +Y direction) intersecting the first direction. The transfer unit **20** has the protrusion (restricting portion) **31a** that restricts the attachment of the fixing unit **40** when the transfer unit **20** is positioned in the first irregular position.

(166) Thus, the user can recognize that the transfer unit **20** has been positioned in the first irregular position (i.e., attached incorrectly) by the restriction of the attachment of the fixing unit **40**. That is, with a simple configuration, it becomes possible to enable the user to recognize incorrect attachment of the transfer unit **20**.

(167) The image forming apparatus **1** further includes the top cover unit (cover member) **50** that is openable and closable with respect to the apparatus body **100**. When the transfer unit **20** is positioned in the second irregular position in the state where the fixing unit **40** is attached to the apparatus body **100**, the closing of the top cover unit **50** is restricted by contact between the protrusions **31a** and the fixing unit **40**. Thus, the user can recognize that the transfer unit **20** has been incorrectly attached by not being able to close the top cover unit **50**.

(168) The unit housing **44** of the fixing unit **40** has the inclined surfaces **46L** and **46R** (first contact portions) that contact the protrusions **31a** when the attachment of the fixing unit **40** is restricted by

the protrusions **31a**, and the restriction surface **45a** (second contact portion) disposed on the side in the detachment direction (A2 direction) of the fixing unit **40** with respect to the inclined surfaces **46L** and **46R**. The protrusions **31a** are elastically deformed by contacting the inclined surfaces **46L** and **46R**, and further contact the restriction surface **45a**, thereby restricting the attachment of the fixing unit **40**. This enables the user to recognize incorrect attachment of the transfer unit **20** without causing damage to the protrusions **31a** or the like.

(169) Since the inclined surfaces **46L** and **46R** (first contact portions) are inclined toward the restriction surface **45a** (second contact portion) (see FIG. 21(B)), the protrusions **31a** can be brought into a state of contacting the restriction surface **45a** while mitigating the impact caused by the contact between the protrusions **31a** and the inclined surfaces **46L** and **46R**. This can enhance the effect of preventing damage to the protrusion **31a** and the like.

(170) The bearing portions **31L** and **31R** of the transfer unit **20** have the fitting portions **31c** (second engagement portions) that can engage the positioning grooves **82L** and **82R** (first engagement portions) of the apparatus body **100** in the attachment direction (second direction) of the transfer unit **20**. The first irregular position of the transfer unit **20** described above is the position where the transfer unit **20** rides onto the positioning grooves **82L** and **82R** in the direction different from the attachment direction of the transfer unit **20** (for example, substantially in the +Z direction). The incorrect attachment of the transfer unit **20** is usually difficult for the user to recognize, but the present embodiment enables the user to recognize the incorrect attachment.

(171) The transfer unit **20** has the guide portions **31b** that are guided by the guide rails **81L** and **81R** (guide members) of the apparatus body **100**, and the above protrusion **31a** is integrally formed with the guide portion **31b**. Therefore, with a simple configuration, it becomes possible to enable the user to recognize the incorrect attachment of the transfer unit **20**.

(172) The protrusions **31a** are provided to protrude from the transfer unit **20** in the attachment direction (substantially in the +Y direction: in the second direction). The protrusions **31a** restrict the attachment of the fixing unit **40** when the transfer unit **20** is located in the first irregular position, and restrict the closing of the top cover unit **50** (cover member) **50** by contacting the fixing unit **40** when the transfer unit **20** is located in the second irregular position. Thus, these functions can be implemented with a simple configuration.

(173) In the above embodiment, the protrusions **31a** of the bearing portions **31L** and **31R** constitute the restricting portions that restrict the attachment of the fixing unit **40** and also restrict the closing of the top cover unit **50**. However, the restricting portion is not limited to the protrusion **31a** in each of the bearing portions **31L** and **31R**, but may be any part of the transfer unit **20**. For example, the protrusion(s) may be formed on the belt frame **26** of the transfer unit **20**.

(174) In the above embodiment, the restricting portions of the transfer unit **20** (the protrusions **31a** of the bearing portions **31L** and **31R**) have the actions to restrict the attachment of the fixing unit **40** and restrict the closing of the top cover unit **50** by contacting the fixing unit **40**, but the restricting portion may have only one of these actions.

(175) In the above embodiment, the bearing portions **31L** and **31R** have the guide portions **31b**, but they are not limited to this example. The transfer unit **20** only needs to have a guide portion that is guided by a guide member in the apparatus body **100**.

(176) The image forming apparatus **1** may have another attachable and detachable belt unit instead of the transfer unit **20**, and may also have another openable and closable cover member instead of the top cover unit **50**.

(177) Although the embodiment has been described above, but the present disclosure is not limited to the above-described embodiment, and various modifications or changes can be made thereto.

(178) The present disclosure can be used for image forming apparatuses (for example, copiers, facsimiles, printers, or multifunction machines) that form images on media using the electrophotographic method.

(179) Various aspects of the present disclosure are collectively described below as appendixes.

(180) (Appendix 1) An image forming apparatus comprising:

(181) an apparatus body; a fixing unit attachable to the apparatus body in a first direction; and a belt unit attachable in an attachment position within the apparatus body in a second direction intersecting the first direction, wherein the belt unit has a restricting portion that restricts attachment of the fixing unit when the belt unit is positioned in a first irregular position different from the attachment position.

(Appendix 2)

(182) The image forming apparatus according to appendix 1, further comprising a cover member provided to be openable and closable with respect to the apparatus body, wherein closing of the cover member is restricted by contact of the restricting portion with the fixing unit when the belt unit is positioned at a second irregular position different from the attachment position in a state where the fixing unit is attached to the apparatus body.

(Appendix 3)

(183) The image forming apparatus according to appendix 2, wherein, when the cover member moves in a closing direction, the belt unit contacts the cover member and is positioned at a position where the closing of the cover member is restricted by contact between the restricting portion and the fixing unit.

(184) (Appendix 4)

(185) The image forming apparatus according to any one of appendixes 1 to 3, wherein a unit housing of the fixing unit includes: a first contact portion that contacts the restricting portion when the attachment of the fixing unit to the apparatus body is restricted by the restricting portion; and a second contact portion disposed in a detachment direction of the fixing unit with respect to the first contact portion, wherein the restricting portion is elastically deformed by contacting the first contact portion and further contacts the second contact portion, thereby restricting the attachment of the fixing unit to the apparatus body.

(Appendix 5)

(186) The image forming apparatus according to appendix 4, wherein the first contact portion has an inclined surface inclined toward the second contact portion.

(187) (Appendix 6)

(188) The image forming apparatus according to any one of claims 1 to 5, wherein the belt unit has a second engagement portion capable of engaging a first engagement portion in the second direction, the first engagement portion being provided in the apparatus body, and wherein the first irregular position of the belt unit is a position where the belt unit rides onto the first engagement portion in a direction different from the second direction.

(Appendix 7)

(189) The image forming apparatus according to any one of appendixes 1 to 6, wherein the apparatus body has a guide member that guides the belt unit in the second direction, wherein the belt unit has a guide portion guided by the guide member, and wherein the restricting portion is formed integrally with the guide portion.

(Appendix 8)

(190) The image forming apparatus according to any one of appendixes 1 to 7, wherein the restricting portion is a plate-shaped member protruding from the belt unit in the second direction.

(191) (Appendix 9)

(192) The image forming apparatus according to any one of appendixes 1 to 8, wherein the belt unit has a belt, a drive roller that drives the belt, and a bearing portion that rotatably supports a shaft of the drive roller, and wherein the restricting portion is provided in the bearing portion.

DESCRIPTION OF REFERENCE CHARACTERS

(193) **1**: image forming apparatus, **10**, **10K**, **10C**, **10M**, **10Y**: process unit (image forming unit), **11**: photosensitive drum (image carrier), **12**: charging roller (charging member), **13**, **13K**, **13C**, **13M**, **13Y**: exposure head (exposure device), **14**: developing roller (developer carrier), **15**: supply roller

(supply member), **20**: transfer unit (belt unit), **21**: transfer roller (transfer member), **22**: transfer belt (belt), **23**: drive roller, **23a**: shaft, **24**: tension roller, **25**: waste toner box, **26**: belt frame, **28L**, **28R**: U-shaped groove, **30**: gripping portion, **31L**, **31R**: bearing portion, **31a**: protrusion (restricting portion), **31b**: guide portion, **31c**: fitting portion (second engagement portion), **31f**: arm, **31h**: support hole, **40**: fixing unit, **44**: unit housing, **45**: inlet opening, **45a**: restriction surface (second contact portion), **46L**, **46R**: inclined surface (first contact portion), **47L**, **47R**: retracted surface, **48**: wall portion, **50**: top cover unit (cover member), **53K**, **53C**, **53M**, **53Y**: toner cartridge (developer storage), **56**: opening, **60**: basket unit, **70**: housing, **72**: accommodation area of the transfer unit **20**, **73**: accommodation area of the fixing unit **40**, **81L**, **81R**: guide rail (guide member), **82L**, **82R**: positioning groove (first engagement portion), **83L**, **83R**: Post, **85**: roller frame, **90**: front cover, **100**: apparatus body, **110**: medium supply section, **120**: medium ejection section, **200**: toner conveyance path.

Claims

1. An image forming apparatus, comprising: an apparatus body having a first engagement portion; a fixing unit attachable to the apparatus body in a first direction, the fixing unit being configured to fix an image to a medium; and a belt unit attachable to the apparatus body at an attachment position therein in a second direction intersecting the first direction, wherein the belt unit has a restricting portion that restricts attachment of the fixing unit when the belt unit is positioned at a first irregular position different from the attachment position, wherein the belt unit has a second engagement portion capable of engaging the first engagement portion in the second direction, and wherein the first irregular position of the belt unit is a position where the belt unit rides onto the first engagement portion in a direction different from the second direction.
2. The image forming apparatus according to claim 1, further comprising a cover member provided to be openable and closable with respect to the apparatus body, wherein closing of the cover member is restricted by contact of the restricting portion with the fixing unit when the belt unit is positioned at a second irregular position different from the attachment position in a state where the fixing unit is attached to the apparatus body.
3. The image forming apparatus according to claim 2, wherein, when the cover member moves in a closing direction, the belt unit contacts the cover member and is positioned at a position where the closing of the cover member is restricted by contact between the restricting portion and the fixing unit.
4. The image forming apparatus according to claim 1, wherein a unit housing of the fixing unit includes: a first contact portion that contacts the restricting portion when the attachment of the fixing unit to the apparatus body is restricted by the restricting portion; and a second contact portion disposed in a detachment direction of the fixing unit with respect to the first contact portion, and wherein the restricting portion is elastically deformed by contacting the first contact portion, and further contacts the second contact portion, thereby restricting the attachment of the fixing unit to the apparatus body.
5. The image forming apparatus according to claim 4, wherein the first contact portion has an inclined surface inclined toward the second contact portion.
6. The image forming apparatus according to claim 1, wherein the apparatus body has a guide member that guides the belt unit in the second direction, wherein the belt unit has a guide portion guided by the guide member, and wherein the restricting portion is formed integrally with the guide portion.
7. The image forming apparatus according to claim 1, wherein the restricting portion is a plate-shaped member protruding from the belt unit in the second direction.
8. The image forming apparatus according to claim 1, wherein the belt unit has a belt, a drive roller that drives the belt, and a bearing portion that rotatably supports a shaft of the drive roller, and

wherein the restricting portion is provided in the bearing portion.

9. The image forming apparatus according to claim 8, wherein the restricting portion protrudes in a direction perpendicular to an axial direction of the drive roller.

10. The image forming apparatus according to claim 8, wherein, when the belt unit and the fixing unit are attached to the apparatus body, the restricting portion protrudes toward the fixing unit.

11. The image forming apparatus according to claim 8, wherein the restricting portion protrudes in a direction away from the belt.

12. The image forming apparatus according to claim 1, further comprising: an image forming unit provided in the apparatus body and being configured to form the image to be transferred to the medium.
